THE DESCENT LEMMA: TENSORING THE EQUALIZER DIAGRAM BY A UNIVERSALLY INJECTIVE MAP KEEPS IT AN EQUALIZER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 6 navigation stubs, one for each labelled equation or result invoked outside the span; none of them replaces a load-bearing step, since the diagram chase is written out inside the span. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The label lemma-descent-lemma is defined only in descent.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Universally injective module maps and the functor C. Same chapter, the construction of the functor C on the category of N with a theta action, used throughout the proof from line 1451. Supplies the objects whose equalizer and coequalizer diagrams are compared.
- Split-coequalizer and equalizer diagrams and their stability. Standard category theory, used at proof lines 1473-1490. Supplies the bookkeeping by which the dashed arrow and the split coequalizer structure are obtained.
- The adjunction allowing tensoring with S inside C. Same chapter, Remark remark-adjunction, invoked at proof line 1480. Supplies the passage from the split coequalizer diagram over R to the one over S.
- Faithfulness of C. Same chapter, Lemma lemma-C-is-faithful, invoked at proof lines 1492-1493. Supplies the final step from the split coequalizer to the equalizer in question.

Source reference. Original equation 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original remark 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *If f is universally injective, then the diagram*
(0.1.1)

$$f_*(M,\theta)\otimes_R S \xrightarrow{\theta\circ(1_M\otimes\delta_0^1)} M\otimes_{S,\delta_1^1} S_2 \underset{1_{M\otimes S_2}\otimes\delta_1^2}{\overset{(\theta\otimes\delta_2^2)\circ(1_M\otimes\delta_0^2)}{\rightrightarrows}} M\otimes_{S,\delta_{12}^1} S_3$$

obtained by tensoring (0) over R with S is an equalizer.

Proof. By Lemma 0 and Remark 0, the map $C(1_N\otimes f): C(N\otimes_R S)\to C(N)$ can be split functorially in N. This gives the upper vertical arrows in the commutative diagram

$$\begin{array}{ccccc}
C(M\otimes_{S,\delta_1^1} S_2) & \underset{C(1_M\otimes\delta_1^1)}{\overset{C(\theta\circ(1_M\otimes\delta_0^1))}{\rightrightarrows}} & C(M) & \longrightarrow & C(f_*(M,\theta)) \\
\downarrow & & \downarrow & & \vdots \\
C(M\otimes_{S,\delta_{12}^1} S_3) & \underset{C(1_{M\otimes S_2}\otimes\delta_1^2)}{\overset{C((\theta\otimes\delta_2^2)\circ(1_M\otimes\delta_0^2))}{\rightrightarrows}} & C(M\otimes_{S,\delta_1^1} S_2) & \xrightarrow{C(\theta\circ(1_M\otimes\delta_0^1))} & C(M) \\
\downarrow & & \downarrow{\scriptstyle C(1_M\otimes\delta_1^1)} & \nearrow & \downarrow \\
C(M\otimes_{S,\delta_1^1} S_2) & \underset{C(1_M\otimes\delta_1^1)}{\overset{C(\theta\circ(1_M\otimes\delta_0^1))}{\rightrightarrows}} & C(M) & \longrightarrow & C(f_*(M,\theta))
\end{array}$$

in which the compositions along the columns are identity morphisms. The second row is the coequalizer diagram (0); this produces the dashed arrow. From the top right square, we obtain auxiliary morphisms $C(f_*(M,\theta))\to C(M)$ and $C(M)\to C(M\otimes_{S,\delta_1^1} S_2)$ which imply that the first row is a split coequalizer diagram. By Remark 0, we may tensor with S inside C to obtain the split coequalizer diagram

$$C(M\otimes_{S,\delta_2^2\circ\delta_1^1} S_3) \underset{C(1_{M\otimes S_2}\otimes\delta_1^2)}{\overset{C((\theta\otimes\delta_2^2)\circ(1_M\otimes\delta_0^2))}{\rightrightarrows}} C(M\otimes_{S,\delta_1^1} S_2) \xrightarrow{C(\theta\circ(1_M\otimes\delta_0^1))} C(f_*(M,\theta)\otimes_R S).$$

By Lemma 0, we conclude (0.1.1) must also be an equalizer. □